

Quantum Beam Spreading

Sehtaj has a thin wall through which she accelerates particles with mass m to velocity v which pass through a small hole of atomic size with diameter D_0 .

Given parameters:

- Mass of particles: m
- Velocity of particles: v
- Initial hole diameter: D_0
- Distance from the hole: L (where $L \gg D_0$)

Task

- Estimate the diameter of the beam D_L at the distance L from the hole.

proposed by Tawhid Bin Omar